

Telescoping Bayesian Inference with Approximate Forward Operators: Corrected Convergence Theory, Multimodality as Loss of Injectivity, and the Cost of Heavy Tails

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Abstract

Multilevel and telescoping methods for Bayesian inverse problems are well developed under ideal conditions. This paper is a *corrected* revision of an earlier manuscript that claimed three results beyond the standard theory. Two of the three claims do not survive scrutiny; the third is provable only in a restricted form, which we give. Specifically, (i) the multilevel regime thresholds are $\gamma = \beta + 2$, not $\gamma = \beta + 1$, and the sub-optimal exponent is $-2 - (\beta + 2 - \gamma)/k$ (the bias is $O(L^{-k})$, so $L \asymp \varepsilon^{-1/k}$); (ii) discretization-induced bimodality *cannot* arise from a linear forward operator with a Gaussian prior — the approximate posterior is then exactly Gaussian at every level — and for the Runge–Kutta 4 surrogate of $\dot{y} = \lambda y$ it is a *coarse-resolution transient* caused by non-injectivity of the numerical map, which disappears once the fold $\lambda = z^* n/T$ leaves the prior’s effective support, rather than persisting at all levels; and (iii) the claimed heavy-tail penalty $\rho(\alpha) = (\alpha - 2)/(\alpha - 1)$ is neither proved nor supported by the earlier paper’s own experiment, which we reproduce and which in fact shows the full rate $-(k + 1)$ for every tail index tested. We state precisely what is proved, what is verified numerically, and what remains open, and we ship a verification driver that regenerates every numeral in this paper.

1 Introduction

1.1 What this paper is

This is the corrected second version of a manuscript (hereafter **V1**) on telescoping Bayesian inference with approximate forward operators. V1’s classical material (telescoping posterior decomposition, expectation convergence, variance decay, credible-interval convergence) is essentially correct and is retained, with repairs. V1’s three advertised novel contributions, however, do not hold up:

1. its multilevel complexity regimes are off by one (§4);
2. its theory of discretization-induced bimodality rests on an invalid Hessian argument and on a characterisation contradicted by its own example, and its empirical “confirmation” is a simulation whose code hard-codes the bimodality it claims to explain (§5);

*Manuscript source, the verification driver, and the two independent audit reports are available at <https://github.com/jpbald93/bayesian-telescoping-v2>.

3. its heavy-tail penalty formula is asserted rather than derived, is attributed to a mis-cited reference, and is not exhibited by its own experiment design (§6).

We therefore *retract* V1's Theorems 9.3 and 10.1 and its Theorem 8.4 as stated, and replace them with the correct statements below. Two of the replacement statements are theorems proved here; the third is explicitly posed as a conjecture with numerical evidence against V1's formula.

1.2 Status of the claims

Claim	Status in V2	Where
Telescoping decomposition	correct (elementary identity)	§3
Expectation convergence $O(n^{-(k+1)})$	correct (unchanged)	Thm 3.3
Variance-difference decay	correct (unchanged)	Thm 3.4
MLMC regimes	corrected (thresholds $\gamma = \beta + 2$)	Thm 4.1
Bimodality from linear F	impossible (new, proved)	Thm 5.1
Bimodality from RK4	transient , not persistent (new)	§5.2
Heavy-tail penalty $\rho(\alpha)$	not proved ; not observed numerically	§6

1.3 Notation

X is a separable Banach space (parameter), Y a separable Hilbert space (observations), $y = F(x) + \eta$ with $\eta \sim N(0, \Gamma)$. μ_0 is a prior, μ^y the exact posterior, and μ_n^y the posterior induced by an approximate forward operator F_n . k is the telescoping order: $\|F_{n+1}(x) - F_n(x)\| \leq C(1 + \|x\|^p)/n^{k+1}$.

2 Preliminaries

We use the Hellinger distance $d_H(\mu, \nu)^2 = \int (\sqrt{d\mu/d\lambda} - \sqrt{d\nu/d\lambda})^2 d\lambda$, which satisfies $d_H \in [0, \sqrt{2}]$ and $d_H = \sqrt{2(1-\rho)}$ with the Bhattacharyya coefficient ρ . (With the alternative normalisation $\frac{1}{2} \int (\cdot)^2$ one has $d_H \in [0, 1]$ and $d_H = \sqrt{1-\rho}$; V1 mixed the two conventions.) For a Gaussian prior satisfying Fernique's theorem there is $a > 0$ with $\int \exp(a\|x\|^2) \mu_0(dx) < \infty$; we shall need the stronger, and separate, *uniform* exponential-moment condition

$$\sup_{n \geq 1} \int_X \exp(a\|x\|^{2p}) \mu_n^y(dx) < \infty \quad (1)$$

with p the polynomial-growth exponent. The potential is $\Phi_n(x; y) = \frac{1}{2} \|\Gamma^{-1/2}(y - F_n(x))\|_Y^2$ and $d\mu_n^y/d\mu_0 = Z_n(y)^{-1} \exp(-\Phi_n(\cdot; y))$.

We record the elementary facts we shall use, with proofs, because V1 got one of them wrong.

Lemma 2.1 (Telescoping tail; corrected). *Let $\alpha > 1$. Then*

$$\sum_{n \geq N} n^{-\alpha} \leq N^{-\alpha} + \int_N^\infty t^{-\alpha} dt = N^{-\alpha} + \frac{N^{1-\alpha}}{\alpha-1} \leq \frac{C_\alpha}{N^{\alpha-1}},$$

and in particular $\sum_{n \geq N} n^{-\alpha} = \Theta(N^{1-\alpha})$ (the tail; note that $\sum_{n < N} n^{-\alpha} = \Theta(1)$ for $\alpha > 1$).

Proof. For decreasing $t \mapsto t^{-\alpha}$ we have $n^{-\alpha} \leq \int_{n-1}^n t^{-\alpha} dt$ for $n \geq 2$, so $\sum_{n \geq N+1} n^{-\alpha} \leq \int_N^\infty t^{-\alpha} dt$, and adding the $n = N$ term gives the first bound; the second is immediate for $N \geq 1$ with $C_\alpha = 1 + 1/(\alpha-1)$. $\square$

Remark 2.2. V1 bounded $\int_{N-1}^{\infty} t^{-\alpha} dt = 1/((\alpha-1)(N-1)^{\alpha-1})$ and then wrote “ $\leq 1/((\alpha-1)N^{\alpha-1})$ ”, which is the wrong direction: the sum exceeds $1/((\alpha-1)N^{\alpha-1})$ (e.g. $\alpha = 2, N = 10$: $0.1052 > 0.1000$). The driver asserts both sides. The conclusion $\Theta(N^{-(\alpha-1)})$ is unaffected; only the displayed step was wrong.

3 Telescoping posteriors

Proposition 3.1 (Increment identity; correct). $\Phi_{n+1}(x; y) - \Phi_n(x; y) = -\langle \Gamma^{-1}(y - F_n(x)), F_{n+1}(x) - F_n(x) \rangle + \frac{1}{2} \|\Gamma^{-1/2}(F_{n+1}(x) - F_n(x))\|^2$.

Proof. Write $a = y - F_n(x)$, $d = F_{n+1}(x) - F_n(x)$, so $y - F_{n+1}(x) = a - d$. Then $\Phi_{n+1} = \frac{1}{2} \langle \Gamma^{-1}(a - d), a - d \rangle = \frac{1}{2} \langle \Gamma^{-1}a, a \rangle - \langle \Gamma^{-1}a, d \rangle + \frac{1}{2} \langle \Gamma^{-1}d, d \rangle$. $\square$

Corollary 3.2. Assume $\|F_n(x)\|_Y \leq C(1 + \|x\|_X^p)$ uniformly in n . Then $|\Phi_{n+1}(x; y) - \Phi_n(x; y)| \leq C(1 + \|x\|_X^{2p})(1 + \|y\|_Y)n^{-(k+1)} + C(1 + \|x\|_X^{2p})n^{-2(k+1)}$.

A standard dominated-convergence argument gives the following (V1’s Theorem 5.2), which we do not repeat in full.

Theorem 3.3 (Posterior convergence of expectations). Under the assumed growth and (1), for every bounded Lipschitz φ there is C with $|\mathbb{E}_{\mu_{n+1}^y}[\varphi] - \mathbb{E}_{\mu_n^y}[\varphi]| \leq Cn^{-(k+1)}$, and hence $|\mathbb{E}_{\mu^y}[\varphi] - \mathbb{E}_{\mu_N^y}[\varphi]| \leq C_1(kN^k)^{-1}$.

Theorem 3.4 (Variance-difference decay). Under the hypotheses of Theorem 3.3, $|\text{Var}_{\mu_{n+1}^y}(\varphi) - \text{Var}_{\mu_n^y}(\varphi)| = O(n^{-(k+1)})$.

Proposition 3.5 (Credible intervals; correct hypotheses). Let φ be bounded with marginal densities p_n under μ_n^y , and let F_n be the distribution function of φ under μ_n^y , with α -quantile $q_\alpha^{(n)}$. Suppose (i) $\inf_n \inf_{|s - q_\alpha| < \delta} p_n(s) \geq c > 0$ for some $\delta > 0$, and (ii) $|F_n(q_\alpha) - \alpha| = O(\delta_n)$, where $q_\alpha = F^{-1}(\alpha)$. Then $|q_\alpha^{(n)} - q_\alpha| = O(\delta_n)$.

Proof. By the mean value theorem applied to F_n , $|q_\alpha^{(n)} - q_\alpha| = |F_n(q_\alpha) - \alpha|/p_n(\xi_n)$ for some ξ_n between the two quantiles. The numerator is $O(\delta_n)$ by (ii); the denominator is $\geq c$ by (i) once $|q_\alpha^{(n)} - q_\alpha| < \delta$. $\square$

Remark 3.6. Hypothesis (ii) is *not* implied by convergence of the first two moments: two log-concave densities with identical mean and variance may have different quantiles. V1’s Corollary 7.2 assumed only moment convergence and is therefore false; what is needed — and what V1’s own Lemma A.1 in fact used — is convergence of the distribution function at the quantile. Note also that (i) is a hypothesis *about the approximate posteriors*; it is automatic when they are log-concave, which §5 shows holds for affine F_n and Gaussian priors, but it can fail in general.

4 Multilevel complexity: corrected regimes

Let $V_n = \text{Var}(\Delta_n) = O(n^{-\gamma})$ and $C_n = O(n^\beta)$, and let the estimator use L levels with optimal allocation $N_n \propto \sqrt{V_n/C_n}$. The standard identity $\text{Cost}(\varepsilon) \asymp \varepsilon^{-2} (\sum_{n < L} \sqrt{V_n C_n})^2$ with $L \asymp \varepsilon^{-1/k}$ gives, since $\sqrt{V_n C_n} \asymp n^{(\beta-\gamma)/2}$,

Theorem 4.1 (MLMC complexity; corrected). Write $s = (\beta - \gamma)/2$ and $L \asymp \varepsilon^{-1/k}$.

1. If $\gamma > \beta + 2$: $\sum_{n < L} n^s = O(1)$ and $\text{Cost}(\varepsilon) = O(\varepsilon^{-2})$.
2. If $\gamma = \beta + 2$: $\sum_{n < L} n^s = \Theta(\log L)$ and $\text{Cost}(\varepsilon) = O(\varepsilon^{-2}(\log \varepsilon)^2)$.
3. If $\gamma < \beta + 2$: $\sum_{n < L} n^s = \Theta(L^{(\beta+2-\gamma)/2})$ and $\text{Cost}(\varepsilon) = O(\varepsilon^{-2-(\beta+2-\gamma)/k})$.

Proof. $\sum_{n < L} n^s$ converges iff $s < -1$, i.e. $\gamma > \beta + 2$; at $s = -1$ it is $\Theta(\log L)$; for $s > -1$ it is $\Theta(L^{s+1}) = \Theta(L^{(\beta+2-\gamma)/2})$. Substituting $L \asymp \varepsilon^{-1/k}$ gives (iii). $\square$

Remark 4.2 (Cutoff: the bias is $O(L^{-k})$, not $O(L^{-(k+1)})$). Theorem 3.3 bounds the *increment* by $Cn^{-(k+1)}$; the level- L bias is the tail sum $\sum_{n \geq L} |\mathbb{E}_{\mu_{n+1}}[\varphi] - \mathbb{E}_{\mu_n}[\varphi]| = O(L^{-k})$ by Lemma 2.1 with $\alpha = k + 1$. Hence $L \asymp \varepsilon^{-1/k}$, not $\varepsilon^{-1/(k+1)}$. (The driver verifies $\sum_{n \geq L} n^{-(k+1)} = \Theta(L^{-k})$.) V1 used $L \asymp \varepsilon^{-1/(k+1)}$, which is inconsistent with its own bias corollary; we correct the cutoff here and correspondingly the exponent in case (iii).

Remark 4.3 (What V1 said, and why it is off by one). V1 asserted the threshold $\gamma > \beta + 1$, the boundary case $\gamma = \beta + 1 \Rightarrow O(\log L)$, and the sub-optimal exponent $-2 - (\beta + 1 - \gamma)/(k + 1)$. Its own derivation contains the slip $(\beta - \gamma)/2 + 1 = (\beta + 1 - \gamma)/2$, which is false: the left side is $(\beta + 2 - \gamma)/2$ (e.g. $\beta = 1, \gamma = 3$: left = 0, right = $-\frac{1}{2}$). With V1's threshold, the partial sums diverge: the driver reports $\sum_{n < 10^6} n^{-3/4} \approx 123$ at the V1 boundary, i.e. V1's " $O(1)$ " case contains divergent sums. The empirical derivation of γ from k (" $\gamma = 2(k + 1)$ ") is in V1 and remains an assumption, not a theorem.

5 Discretization error and multimodality

V1's Theorem 9.3 claimed that "oscillatory error" (RK4) causes bimodality whereas "sign-definite error with $k \geq 2$ " prevents it, and offered a Hessian argument. That argument is invalid: the Hessian of $-\log(d\mu_n^y/d\mu_0)$ omits the prior curvature, and the middle term below is missing from V1. Let H_n be the Hessian of the negative logarithm of the *density* of μ_n^y with respect to Lebesgue measure (prior included):

$$H_n(x) = \nabla F_n(x)^\top \Gamma^{-1} \nabla F_n(x) - \langle \Gamma^{-1}(y - F_n(x)), \nabla^2 F_n(x) \rangle + \text{Prior}^{-1}, \quad (2)$$

and V1 dropped the middle term, which is indefinite and $O(1)$ in n . Two of V1's claims also contradict its own example: RK4 is a *fourth*-order method (V1 labels it $k = 1$), and the "odd/even-order" dichotomy is not the same as the " $k < 2$ / $k \geq 2$ " dichotomy.

We replace Theorem 9.3 with two correct statements.

5.1 An obstruction: linear forward operators cannot produce bimodality

Theorem 5.1 (No bimodality for linear F ; new). *Let $F_n(x) = A_n x + b_n$ be affine and let the prior be Gaussian, $\mu_0 = N(m_0, \Sigma_0)$, with Gaussian noise. Then for every n the approximate posterior μ_n^y is Gaussian, hence log-concave and unimodal; in particular no choice of discretisation of a linear forward operator can induce a bimodal posterior at any level.*

Proof. $\Gamma^{-1/2}(y - A_n x - b_n)$ is affine in x , so $-\Phi_n$ is a concave quadratic in x and $\exp(-\Phi_n)d\mu_0$ is (up to a constant) a Gaussian density; hence μ_n^y is Gaussian. Gaussian densities are strictly log-concave when $\Sigma_n \succ 0$. $\square$

Remark 5.2. This is the sharpest correction to V1. It shows the pathology *must* come from non-linearity: in the O’Brien et al. setting the ODE $\dot{y} = \lambda y$ is linear in y but its numerical solution is nonlinear in the *parameter* λ . V1’s “sign-definite $\Rightarrow$ unimodal” criterion is simply the linear case in disguise.

5.2 The mechanism: non-injectivity of the numerical map

Consider $y = \varphi(\lambda) + \eta$, $\eta \sim N(0, \sigma^2)$, with the exact map $\varphi(\lambda) = e^{\lambda T}$ and its RK4 surrogate with n steps of size $h = T/n$,

$$\varphi_n^{\text{RK4}}(\lambda) = R(\lambda h)^n, \quad R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

e^z is strictly increasing, but R is *not*: $R'(z) = 1 + z + z^2/2 + z^3/6$ has a root $z^* = -1.5960716 \dots \in (-2, -1)$ at which R attains an interior minimum $R(z^*) = 0.2703948$. Consequently $\lambda \mapsto R(\lambda h)^n$ is non-injective, with its fold at

$$\lambda_{\text{fold}}(n) = \frac{z^* n}{T}, \quad (3)$$

while the exact map $e^{\lambda T}$ is strictly increasing. The fold is present for *every* n and simply moves out to $|\lambda| = |z^*|n/T$; what decides whether the posterior is bimodal is whether $\lambda_{\text{fold}}(n)$ lies in the region carrying posterior mass, i.e. the prior’s effective support $[m_0 - 3\sigma_0, m_0 + 3\sigma_0]$. For the tables below ($m_0 = -1$, $\sigma_0 = 0.5$, $T = 1$) this gives

$$|z^*| \frac{n}{T} \lesssim |m_0| + 3\sigma_0 = 2.5 \quad \Longleftrightarrow \quad n \lesssim 1.57,$$

so observable-mass bimodality is confined to $n = 1$; for larger n the fold lies outside the prior-supported region and the surviving secondary modes carry negligible mass (quantified below).

The trapezoidal rule is different: $T(z) = (1 + z/2)/(1 - z/2)$ has $T'(z) = 1/(1 - z/2)^2 > 0$, so its surrogate is strictly increasing on the physical branch $z < 2$ and cannot fold there. (It has a pole at $z = 2$ and, for even n , a second preimage in the far tail $\lambda h > 2$, but with negligible mass in the configurations considered.)

n	exact	RK4	trapezoidal
1	1 mode	2 modes: -1.02 and -2.02 (mass 0.11)	1 mode
2	1 mode	2 modes: -1.00 and -4.71 (mass 2×10^{-13})	1 mode
4	1 mode	1 mode	1 mode
8	1 mode	1 mode	1 mode
16	1 mode	1 mode	1 mode
64	1 mode	1 mode	1 mode

Table 1. All local maxima of the posterior for λ ($T = 1$, prior $N(-1, 0.5^2)$, $\sigma = 0.05$, data $y = e^{-1}$), computed by the shipped driver with *no* height threshold; “mass” is the relative *peak height* of the secondary mode (the integrated relative mass at $n = 1$ is 0.094). RK4 has a genuinely observable second mode (mass $O(10^{-1})$) only at the coarsest level; by $n = 2$ the secondary mode is already exponentially suppressed, and it lies outside the prior-supported region for $n \geq 4$.

Remark 5.3 (Correction of V1’s Experiment 5). V1 reported that RK4 “remains bimodal even at $n = 64$ ”. Table 1 shows the opposite: an observable-mass fold exists only while (3) fails, so the RK4 secondary mode’s mass drops from ≈ 0.11 at $n = 1$ to $\approx 2 \times 10^{-13}$ at $n = 2$ and is confined to the far prior tail thereafter. Moreover V1’s published code for this experiment does not integrate

any ODE: it hard-codes a bimodal *mixture prior* $0.5N(-2, 1) + 0.5N(2, 1)$ and “refines” by dividing the observation noise by $\sqrt{n}$. That experiment therefore says nothing about discretisation error, and V1’s Table 8 has no reproducible source. We supply a genuine implementation in the driver.

Remark 5.4 (A correct diagnostic). Equation (3) gives practitioners a check that V1’s “sign of $F_{n+1} - F_n$ ” does not: compare the fold location $\lambda_{\text{fold}}(n) = z^*n/T$ with the prior’s effective support $[m_0 \pm 3\sigma_0]$. Bimodality is possible only while the fold lies in that range, i.e. $n \lesssim (|m_0| + 3\sigma_0)T/|z^*|$; it is not a fine-grid phenomenon.

6 Heavy-tailed priors: what is proved, and what is not

V1’s Theorem 10.1 claimed a sharp rate $\rho(\alpha)(k+1)$ with $\rho(\alpha) = (\alpha - 2)/(\alpha - 1)$ for priors with polynomial tails $\sim \|x\|^{-\alpha}$. The proof is a truncation argument that does not close (the “ $2p$ term cancels in the optimization” step is asserted, never performed), the matching lower bound is announced rather than derived, and the reference supporting it is mis-cited: V1’s [38] gives DOI 10.1214/24-AOS2356, which resolves to *Stankewitz et al.*, “Early stopping for L^2 -boosting in high-dimensional linear models”; the intended work is Agapiou–Castillo, *Heavy-tailed Bayesian nonparametric adaptation*, Ann. Statist. **52**(4) (2024), 1433–1459, DOI 10.1214/24-AOS2397.

What *is* true is the following sufficient condition, which is what V1’s own Lemma 6.2 in fact uses.

Proposition 6.1 (Sufficient condition). *If the uniform exponential-moment condition (1) holds, then the conclusion of Theorem 3.3 holds with the full rate $O(n^{-(k+1)})$; no degradation occurs.*

Proof. With $|\Delta\Phi_n| \leq C(1 + \|x\|^{2p})n^{-(k+1)}$ (Corollary 3.2) and Lemma $|e^{-t} - 1| \leq |t|e^{|t|}$, one has $\mathbb{E}_{\mu_n}[|e^{-\Delta\Phi_n} - 1|] \leq Cn^{-(k+1)}\mathbb{E}_{\mu_n}[(1 + \|x\|^{2p})e^{C(1+\|x\|^{2p})n^{-(k+1)}}]$, which is $O(n^{-(k+1)})$ uniformly provided (1) holds. The normalisation ratio $Z_n/Z_{n+1} = 1 + O(n^{-(k+1)})$ follows the same way. Combining with the two-part split of V1’s proof of Theorem 6.1 gives the claim. $\square$

Remark 6.2 (The condition is sufficient, not necessary — and V1’s experiment shows no penalty). Condition (1) fails for *any* polynomial-tailed posterior (indeed for any non-Gaussian tail), so it cannot by itself establish degradation; and it turns out not to force it either. We reproduced V1’s own Experiment 6 (which uses $F_n(\theta) = \theta(1 - n^{-2})$, so $k = 2$) with priors Gaussian, t_{10} , t_5 , t_3 , t_2 , computing the *consecutive*-level increment $|\mathbb{E}_{\mu_n}[\theta] - \mathbb{E}_{\mu_{n+1}}[\theta]|$:

prior	Gaussian	t_{10}	t_5	t_3	t_2
measured slope	−2.97	−2.97	−2.97	−2.97	−2.97
V1’s $\rho(\alpha)(k+1)$	−3.000	−2.667	−2.250	−1.500	0.000

The measured rate is $-(k+1)$ in every case; there is no degradation, and V1’s Table 9 (which reported $-2.61, -2.19, -1.47$ for the same priors) is not reproducible from its own model. What goes wrong in V1 is the inference “the sufficient condition fails $\Rightarrow$ the rate degrades”.

Conjecture 6.3 (Open problem). Characterise the exact degradation of the telescoping rate under heavy-tailed priors: for which forward-approximation families and which tail indices does $|\mathbb{E}_{\mu_n}[\varphi] - \mathbb{E}_{\mu_{n+1}}[\varphi]|$ fail to be $\Theta(n^{-(k+1)})$? We expect a genuine penalty to require the increment functional to be unbounded in the tail (so that $\mathbb{E}_{\mu_n}[\|x\|^{2p}]$ is not uniformly bounded), which the model above does not satisfy; but we have neither a proof of a positive rate in general nor a counterexample to one. V1’s formula $\rho(\alpha) = (\alpha - 2)/(\alpha - 1)$ remains unproved and is not supported by numerics.

7 An extension, with corrected cost accounting

V1 proposed a telescoping Kalman filter (TKF) refining the transition model A_n in time and claimed cost $O(T^{3/2})$ versus $O(T^2)$ for a fixed-model filter. That accounting is wrong: with the schedule $n_t = \lceil \sqrt{t} \rceil$, the per-step cost is $n_t^2 \asymp t$, so $\sum_{t=1}^T n_t^2 = \Theta(T^2)$ — the same as $T \cdot n_T^2$. The driver reports $\sum_{t \leq 10^4} n_t^2 = 5.07 \times 10^7$ against $T^{3/2} = 10^6$. There is therefore no asymptotic advantage; any gain is a constant-factor/early-transient effect, not a complexity improvement. We retain the TKF as a heuristic with no asymptotic claim.

8 Discussion

The corrected picture is smaller but sound. Telescoping Bayesian inference gives, under clean exponential moments, the rate $O(n^{-(k+1)})$ for posterior expectation differences (Theorem 3.3), the same rate for variance differences (Theorem 3.4), and the standard multilevel cost with the correct threshold $\gamma = \beta + 2$ (Theorem 4.1). The two “non-ideal” phenomena that motivated V1 behave quite differently from what V1 claimed: discretisation-induced bimodality is impossible for linear forward operators (Theorem 5.1) and, when it does occur, is a coarse-resolution transient tied to non-injectivity of the numerical map (§5.2); and the heavy-tail “penalty” is, at least in the model that was supposed to exhibit it, absent (§6). Both are genuine open problems, now stated honestly.

Reproducibility. Every number in this paper is regenerated by `code/verify_bt_v2.py`, which asserts every claimed sign and threshold and exits nonzero on failure; it reports **ALL ASSERTIONS PASS**. The manuscript, the driver and the two independent vendor audit reports are archived at <https://github.com/jpbald93/bayesian-telescoping-v2>. Unlike V1, we claim no reproduction beyond these artifacts, and we cite no experiment we did not run.

Change log relative to V1

- **Retracted:** V1 Thm 9.3 (bimodality characterisation) — invalid Hessian argument; replaced by Thm 5.1 and §5.2.
- **Retracted:** V1 Thm 10.1 (heavy-tail penalty) — unproved and unobserved; replaced by Prop 6.1 and Conjecture 6.3 (see §6).
- **Corrected:** V1 Thm 8.4 thresholds $\beta + 1 \rightarrow \beta + 2$ and exponent $-2 - (\beta + 1 - \gamma)/(k + 1) \rightarrow -2 - (\beta + 2 - \gamma)/k$.
- **Corrected:** telescope tail lemma integral step (Lemma 2.1).
- **Corrected:** TKF cost $O(T^{3/2}) \rightarrow \Theta(T^2)$.
- **Corrected:** reference for heavy-tailed priors (DOI/venue/authors); the multimodal reference in V1 cited arXiv:0909.2126, which is a different 2009 paper (the intended work is O’Brien, Moores et al., arXiv:2502.11510, 2025).
- **Removed:** all numerical tables not reproducible from a shipped script, and the claim of a public reproduction repository.

Code and data availability

The complete reproduction package — L^AT_EX source, compiled PDF, the verification driver `code/verify_bt_v2.py` together with its recorded output, and the two independent vendor audit reports — is archived at <https://github.com/jpbald93/bayesian-telescoping-v2>. Running `python3 verify_bt_v2.py` (NumPy + mpmath) regenerates every numeral in this paper and exits nonzero if any asserted sign or threshold fails. No table, figure or numerical claim in this paper depends on code or data outside that package.

Acknowledgments

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A Verification driver

`code/verify_bt_v2.py` (Python; NumPy + mpmath; <https://github.com/jpbald93/bayesian-telescoping-v2>) regenerates every numeral above and asserts: the tail bound (Lemma 2.1); the corrected MLMC regimes (Theorem 4.1), including the divergence at V_1 ’s threshold; monotonicity of e^z and $T(z)$ and the interior minimum of R ; the RK4/trapezoidal mode counts of Table 1; and the heavy-tail slopes of §6. It exits nonzero on any failed assertion and prints ALL ASSERTIONS PASS.